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HTJ

HALF YEARLY EXAMINATION - 2025 THANJAVUR-DISTRICT

12 - Std PHYSICS

Time : 3.00 Hrs Marks : 70

I **Note : (i) Answer all the questions.** **15x1=15**
(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer.

1. Two points A and B are maintained at a potential of 7V and -4V respectively. The work done in moving 50 electrons from A to B is
a) 8.80×10^{-17}J b) -8.80×10^{-17}J c) 4.40×10^{-17}J d) 5.80×10^{-17}J

2. At infinity, the electrostatic potential is
a) infinity b) maximum c) minimum d) zero

3. A carbon resistor of (47 ± 4.7) kΩ to be marked with rings of different colours for its identification. The colour code sequence will be
a) Yellow – Green – Violet – Gold b) Yellow – Violet – Orange – Silver
c) Violet – Yellow – Orange – Silver d) Green – Orange – Violet – Gold

4. The value of Bohr magneton μ_B is
a) 9.27×10^{24} Am b) 9.27×10^{24} Am^{-2} c) 9.27×10^{24} Am^{-1} d) 9.27×10^{-24} Am^2

5. In an electrical circuit, R, L, C and AC voltage source are all connected in series. When L is removed from the circuit, the phase difference between the voltage and current in the circuit is $\pi/3$. Instead, if C is removed from the circuit, the phase difference is again $\pi/3$. The power factor of the circuit is
a) $\frac{1}{2}$ b) $\frac{1}{\sqrt{2}}$ c) 1 d) $\frac{\sqrt{3}}{2}$

6. inductor is connected to a capacitor of capacitance C. The value of C in order to impart maximum power at 50Hz is
a) 50μF b) 0.5μF c) 500μF d) 5μF

7. In an electromagnetic wave travelling in free space the rms value of the electric field is $3Vm^{-1}$. The peak value of the magnetic field is
a) 1.414×10^{-8} T b) 1.0×10^{-8} T c) 2.828×10^{-8} T d) 2.0×10^{-8} T

8. An object is placed in front of a convex mirror of focal length of f and the maximum and minimum distance of an object from the mirror such that the image formed is real and magnified.
a) 2f and c b) c and ∞ c) f and O d) None of these

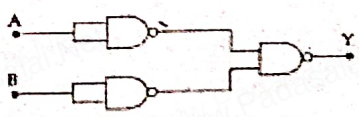
9. Stars twinkle due to
a) reflection b) total internal reflection c) refraction d) polarisation

10. If a beam of unpolarised light is incident on a reflecting glass surface at an angle of 57.5^o, then the angle between the reflected and refracted beam will be
a) 45° b) 60° c) 90° d) 30°

11. When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential is V/4. The threshold wavelength for the metallic surface is
a) 4 λ b) 5 λ c) 5/2 λ d) 3 λ

12. The ratio of the wavelengths of the radiation emitted for the transition from n=2 to n=1 in Li^{++}, He^{+} and H is
a) 1:2:3 b) 1:4:9 c) 3:2:1 d) 4:9:36

13. The barrier potential of a silicon diode is approximately,
a) 0.7V b) 0.3 V c) 2.0 V d) 2.2 V

14. The following arrangement performs the logic function of ..... gate


a) AND b) OR
c) NAND d) EXOR

15. The technology used for stopping the brain from processing pain is
a) Precision medicine b) Wireless brain sensor c) Virtual reality d) Radiology

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II Answer any six Questions. Qn.No.24 is compulsory. **6 x 2 = 12**

16. Define 'electric dipole'. Give the expression for the magnitude of its electric dipole moment and the direction.
17. What is meant by wattles current?
18. State Ampere's circuital law.
19. Why does sky appear blue?
20. Two light sources of equal amplitudes interfere with each other. Calculate the ratio of maximum and minimum intensities.
21. How will you define threshold frequency?
22. Define Curie.
23. What do you meant by doping?
24. If an electric field of magnitude $570NC^{-1}$, is applied in the copper wire, find the acceleration experienced by the electron.

III Answer any six Questions. Qn.No.33 is compulsory. **6 x 3 = 18**

25. Obtain the expression for energy stored in the parallel plate capacitor.
26. State Kirchoff's current rule and Voltage rule.
27. Discuss the conversion of galvanometer into a voltmeter.
28. Find the impedance of a series RLC circuit if the inductive reactance, capacitive reactance and resistance are 184Ω, 144Ω and 30Ω respectively. Also calculate the phase angle between voltage and current.
29. Write down Maxwell equations in integral form.
30. Derive the relation between f and R for a spherical mirror.
31. What is Zener diode? Mention any two uses of Zener diode.
32. Write the properties of neutrino.
33. A proton and an electron have same de Broglie wavelength. Which of them moves faster and which possesses more Kinetic energy?

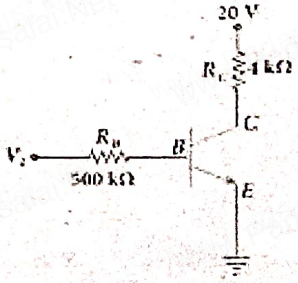
IV Answer all the Questions : **5 x 5 = 25**

34. a) Obtain lens maker's formula and mention its significance. ***(OR)***
    b) Explain in detail the construction and working of a Van de Graaff generator.
35. a) Discuss the diffraction at single slit and obtain the condition for n^{th} minimum. ***(OR)***
    b) Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current using Biot-Savart's law.
36. a) Explain the construction and workings of transformer and define its efficiency. ***(OR)***
    b) Obtain Einstein's photo electric equation with necessary explanation.
37. a) Explain the determination of the internal resistance of a cell using voltmeter. ***(OR)***
    b) Explain the J.J.Thomson experiment to determine the specific charge of electron.
38. a) (i) Write down the properties of electromagnetic waves.
    (ii) A transmitter consists of LC circuit with an inductance of 1μH and a capacitance of 1μF. What is the wavelength of the electromagnetic waves it emits?

    (OR)

    b) (i) Draw the circuit diagram of NPN transistor in common emitter configuration. (ii) In the circuit shown in the figure, the input voltage V_i is 20V, V_{BE} = 0 V and V_{CE} = 0 V.

    What are the values of I_C, I_B and β?



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